DAY - 3 SEAT NUMBER | | | | | | |

2011	II	26	1500	V – 368	(E)

COMPUTER SCIENCE
PAPER - II (D-9)

Time : 3 Hours	4 Pages	Max. Marks : 50

Instructions : *(1) All questions are compulsory.*
(2) Figures to the right indicate full marks.
(3) Draw a neat diagram wherever necessary.
(4) Due credit will be given for any correct logic and appropriate comment.

1. (A) Select correct alternatives and rewrite the following :

(a) Bus Topologies are best suited for networks that use _____ Access Methods. 1

(i) Contention Based
(ii) Token Passing
(iii) Polling
(iv) None of these

(b) In 8051, 22 special function registers occupy memory space from _____. 1

(i) 08H to F8H
(ii) 80H to F8H
(iii) 80 to 8FH
(iv) None of those

(c) During PUSH instruction of 8085, the stack pointer _____. 1

(i) Increment by 1
(ii) Increment by 2
(iii) Decrement by 1
(iv) Decrement by 2

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(d) PSW is a combination of _____ registers. 1

(i) M and F

(ii) H and F

(iii) L and F

(iv) A and F

(B) Answer **any two** of the following :

(a) Explain function of the following Pin in 8085 Microprocessor : 3

(i) HOLD

(ii) READY

(iii) INTA

(b) State RAM and ROM size of the following Microcontrollers 8048, 8049, 8050. 3

(c) Explain the following instructions of 8085 Microprocessor : 3

(i) NOP

(ii) XRA Reg.

(iii) LHLD Addr

2. (A) Answer **any two** of the following :

(a) Explain the following Registers of 8085 Microprocessor : 3

(i) Stack Pointer

(ii) Program Counter

(iii) Instruction Register

(b) What is Protocol ? Explain TCP/IP. 3

(c) What are the different ways of clearing Accumulator (A = 00H) in single instruction ? 3

(B) Attempt **any one** of the following :

(a) Define the following terms with suitable diagram : 4

(i) I. State

(ii) Machine Cycle

(iii) Instruction Cycle

(b) Explain main features of a Pentium Processor. 4

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3. (A) Answer **any two** of the following :

(a) Compare between UTP and Fiber Optic Cable. 3

(b) Explain ALU of Generic Microprocessor. 3

(c) Differentiate between Microcontrollers 8051 and 8052. 3

(B) Answer **any one** from the following :

(a) State any eight features of Microcontroller 8051. 4

(b) Explain Flag Register of 8085 Microprocessor with example. 4

4. (A) Answer **any two** of the following :

(a) Differentiate between LAN and WAN. 3

(b) Explain any three addressing modes of 8085 Microprocessor with example. 3

(c) If Accumulator contains data ABH and C register contains EFH, what will be the contents of accumulator after execution of the following instructions ? 3

(i) ADD C

(ii) SUB C

(iii) ORA C

(B) Answer **any one** of the following :

(a) What is Topology ? Explain BUS and STAR Topologies. 4

(b) Write functions of each of the following devices in short : 4

(i) Modem

(ii) HUB

(iii) Router

(iv) Repeater

5. Answer **any two** of the following :

(a) Write an Assembly Language Program to odd all odd numbers stored in memory block of 10 locations starting from 2000H. Store two byte sum at memory location starting from 3000H. 5

(b) Write an Assembly Language Program to separate nibbles of a number stored at memory location 2000H. Multiply separated nibbles and store result at memory location 4000H. 5

(c) Write a program in Assembly Language to transfer a block of data from 1050 to 1059 to memory location whose starting address is 1070H. 5

OR

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5. Answer **any two** of the following :

(a) Write an Assembly Language Program to count number of even data bytes occurring in a block starting from memory location C030H to C039H. Store result at the memory location C040H. 5

(b) Write an Assembly Language Program to exchange position of digit of number stored at C040H. Multiply original number with the exchanged number, the result to be stored at memory location starting from C041H onwards. 5

(c) Write an Assembly Language Program to add two 16 bit numbers. The numbers are stored at memory location C030H and C031H, and the second number stored at C032H and C033H. Store result at memory location C034H and C035H. Store final carry at C036H. 5

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